

Cytogenetic Assessment of Ionizing Exposure in *Allium cepa* Using Micronucleus Frequency: A Comparative Study Between Antarctic and Control Conditions



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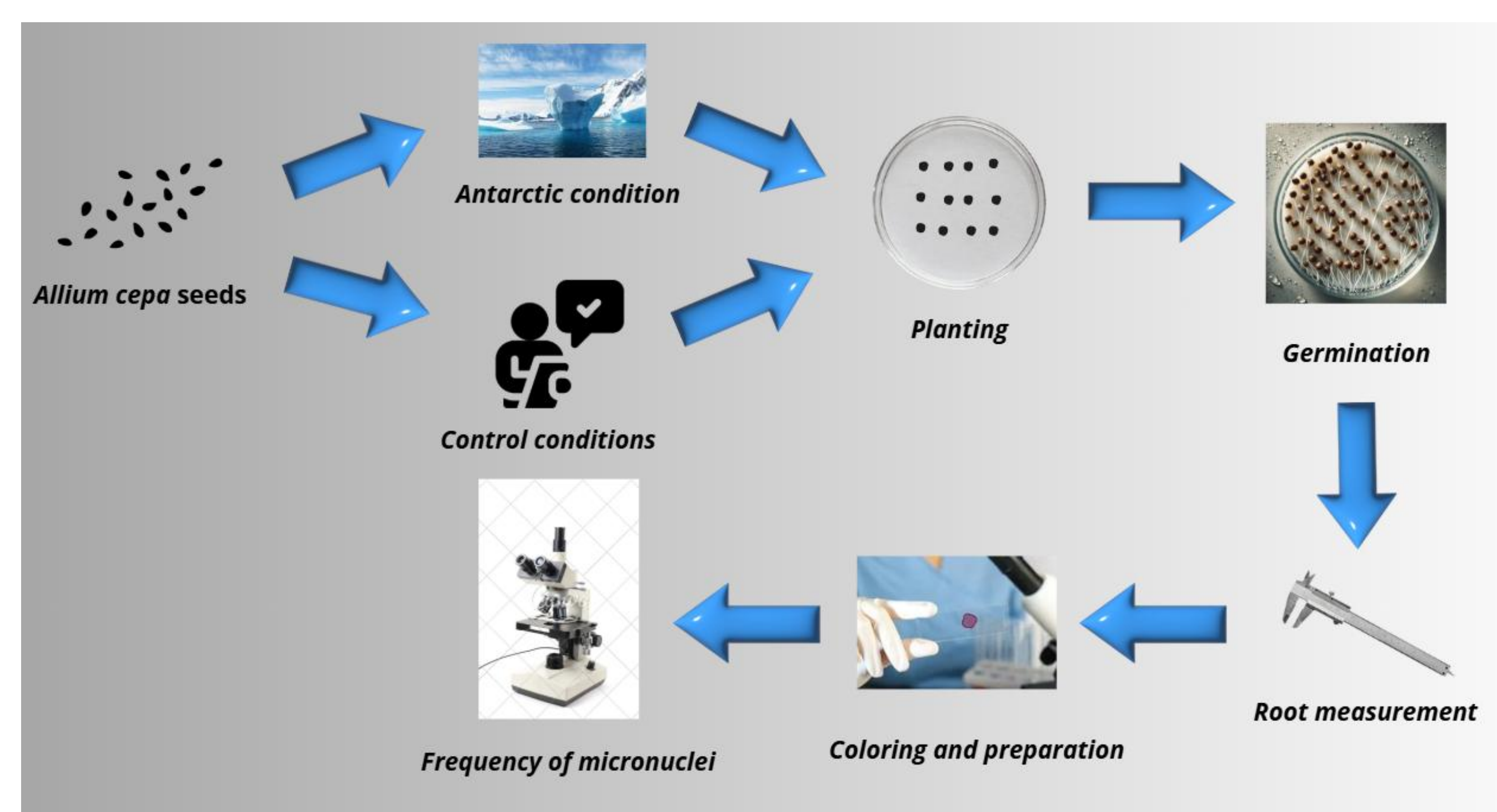
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1. INTRODUCTION

Ionizing radiation in space and polar environments demands reliable monitoring tools beyond physical dosimeters, which cannot capture biological variability. The *Allium cepa* micronucleus test is a low-cost and sensitive biodosimetric method, but its performance under low-dose exposures in dry seeds remains unclear.

This study evaluated dry onion seeds exposed in Antarctica, both inside and outside an anthropomorphic phantom, compared with controls in Brazil. Cytogenetic markers were analyzed and correlated with physical dosimetry to assess the feasibility of using dry seeds as practical biodosimeters for low-dose environmental radiation.

2. MATERIALS AND METHODS



3. RESULTS

○ Macroscopy analysis

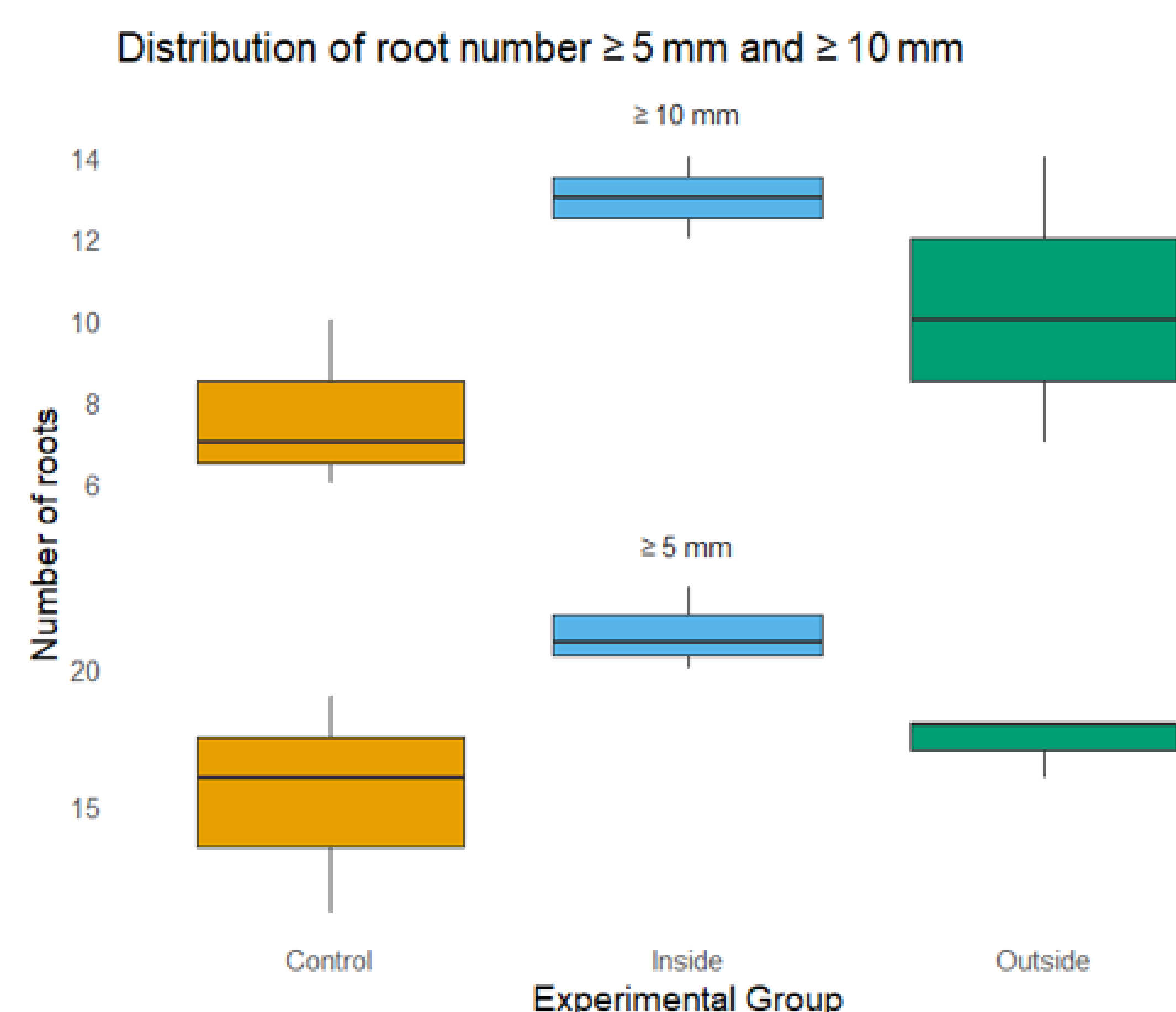


Figure 1: Distribution of the number of roots of *Allium cepa* with lengths ≥ 5 mm and ≥ 10 mm in the Control, Inside and Outside groups of the experimental environment.

○ Microscopy analysis

Treatment	Number of cells in division	MI (%)	Phase Index			
			P	M	A	T
Control	10,6 $\pm$ 3,01	1,06% $\pm$ 0,30%	2,4 $\pm$ 1,62	3,2 $\pm$ 2,23	2,0 $\pm$ 1,41	3,0 $\pm$ 1,79
Inside	19,8 $\pm$ 6,76	1,98% $\pm$ 0,68%	5,6 $\pm$ 2,50	3,4 $\pm$ 1,50	4,6 $\pm$ 3,26	6,2 $\pm$ 2,14
Outside	15,6 $\pm$ 4,22	1,56% $\pm$ 0,42%	2,4 $\pm$ 2,33	4,0 $\pm$ 2,28	3,8 $\pm$ 0,75	5,4 $\pm$ 1,96

Table 1: Mitosis in meristematic cells of *Allium cepa*: number of dividing cells, mitotic index (MI) and percentage distribution of mitotic phases (P: prophase; M: metaphase; A: anaphase; T: telophase) in the different experimental groups.

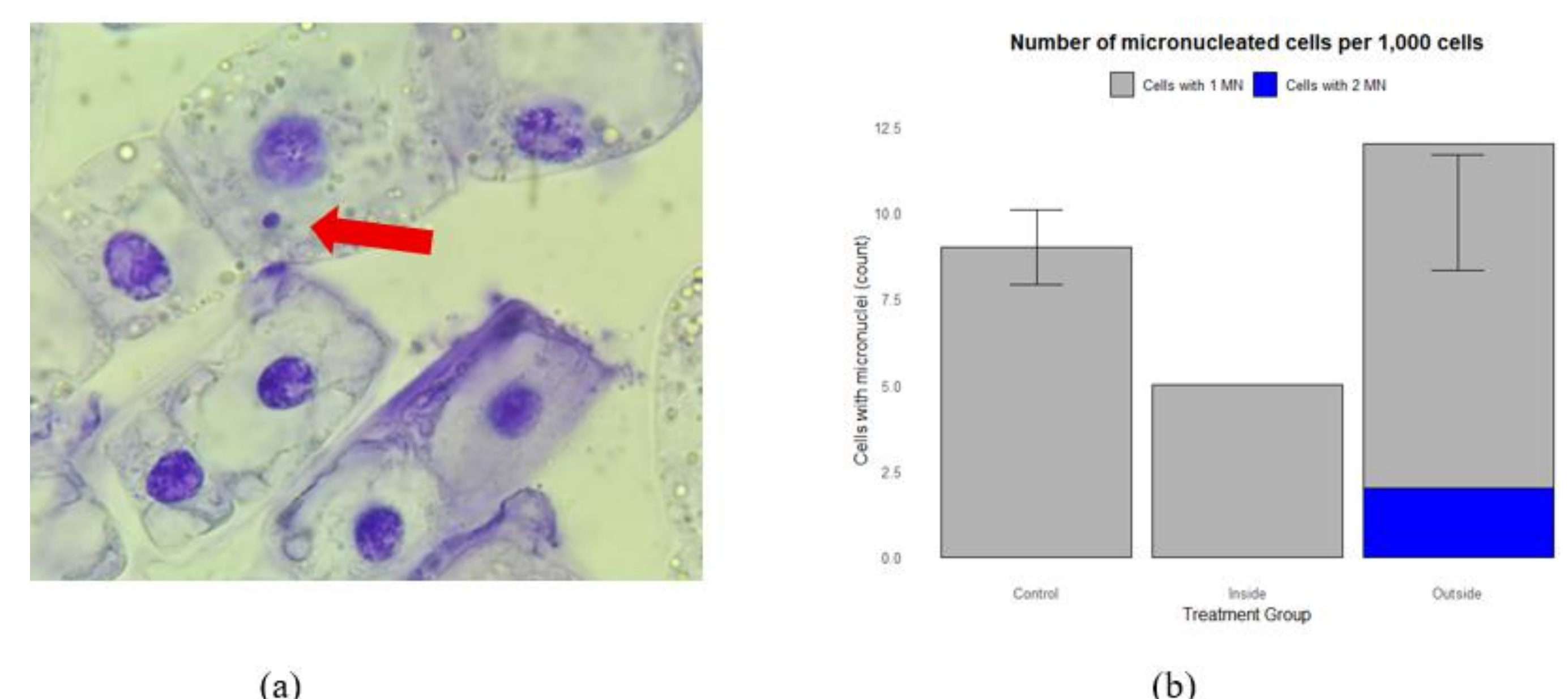


Figure 2: (a) Image obtained by optical microscopy of meristematic cells of *Allium cepa*, showing micronuclei (red arrows). (b) Chart showing the number of cells containing 1, 2, or 3 micronuclei per 1,000 cells, in the Control, In, and Out groups. Cells with 2 MN are highlighted in blue. Error bars represent the standard deviation.

4. CONCLUSIONS

Exposure of dry onion seeds to low-dose ionizing radiation (~ 49 μ Sv), comparable to levels found on flights or in Antarctica, produced no detectable cytogenetic changes by micronucleus assay. This absence of effects reflects both the low radiation dose and the reduced sensitivity of quiescent tissues with low metabolic activity and higher resistance to genetic damage.

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